

Seating before locking a single axis compression pad

Abstract

A pivoting compression pad can accommodate a sloping coupon during seating, yet remain free to move during the measurement ramp. This teaching study compares three timings of an existing angular clamp: locked before approach (E), free throughout (F), and locked after seating at 10 N (L). In constructed pitch-slope records, L gives a force coefficient of variation of 2.3%, compared with 7.2% for E and 2.1% for F. Its median lateral drift is 19 micrometres, versus 125 for F, at a seating-time cost of 32 rather than 17 seconds. Flat coupons show no repeatability advantage from L, and a roll slope remains poorly accommodated. The records support a conditional procedural tradeoff, not a new joint or a demonstration of accurate material properties.

Introduction

A rigid horizontal pad may first meet a sloping coupon near one edge. Allowing the pad to pitch can accommodate that slope before compression, but freedom useful during approach need not be desirable during the measurement ramp. The existing fixture offers both a transverse pivot pin and a friction clamp. The relevant choice is therefore when to constrain the pad angle: before contact, never during loading, or only after seating. This comparison asks whether the last sequence can retain low reseating variation while limiting lateral motion.

The change is operational. The compression frame, sensors, rectangular contact pad, pivot, clamp and lower platen already existed; no new joint or feedback controller was built. Figure 1 compares their states under the three procedures. A pitch slope tests the available angular freedom, whereas a perpendicular roll slope tests its limit. A flat coupon asks whether the extra locking action is useful when angular accommodation is unnecessary. Force calibration, surface finish, sensor sampling and analysis software remain unchanged across modes.

Method

In E, the pad starts horizontal and is clamped before approach to 10 N. In F, the clamp stays open through approach and the subsequent ramp. In L, the open pad first seats at 10 N; the operator holds that preload, acknowledges the hold and tightens the same clamp without lifting the pad. A missing acknowledgement aborts the run, although none is missing here. After the prescribed approach or lock sequence, displacement is zeroed and increased by 0.20 mm. The clamp is released only after unloading.

The pin axis is perpendicular to the side-view plane, permitting pitch but no second-axis roll. Coupons have planar tops that are flat, tilted 1 degree in pitch or tilted 1 degree in roll. The 10 N preload and 0.20 mm increment are inherited teaching settings, not optimized values. L adds an operator action during the hold; lock force is neither measured nor automatically corrected. A settled pose illustrates permitted alignment, not a measured contact-pressure distribution.

Force repeatability at the end of the fixed ramp is calculated from the sample standard deviation s and mean μ across the eight reseatings: $CV = 100 s / \mu$. The value is reported in percent. It is not a measure of force accuracy.

Each row summarizes eight remove-and-reseat cycles of one coupon in that condition, not eight independent specimens. The same coupon geometry and synthetic inputs are used across modes. **The numbers are constructed demonstration values and have no uncertainty intervals.**

Drift is the median magnitude of lateral pad displacement during the ramp, not a rotation angle or pressure metric. Seating time is the median interval from approach initiation to ramp start, including any hold and locking action. It excludes unloading and coupon preparation; full test duration was not measured. Mode order was not randomized. The notebook supplies no raw force samples, quantitative angle trace or pressure map, limiting both uncertainty analysis and attribution of the observed differences.

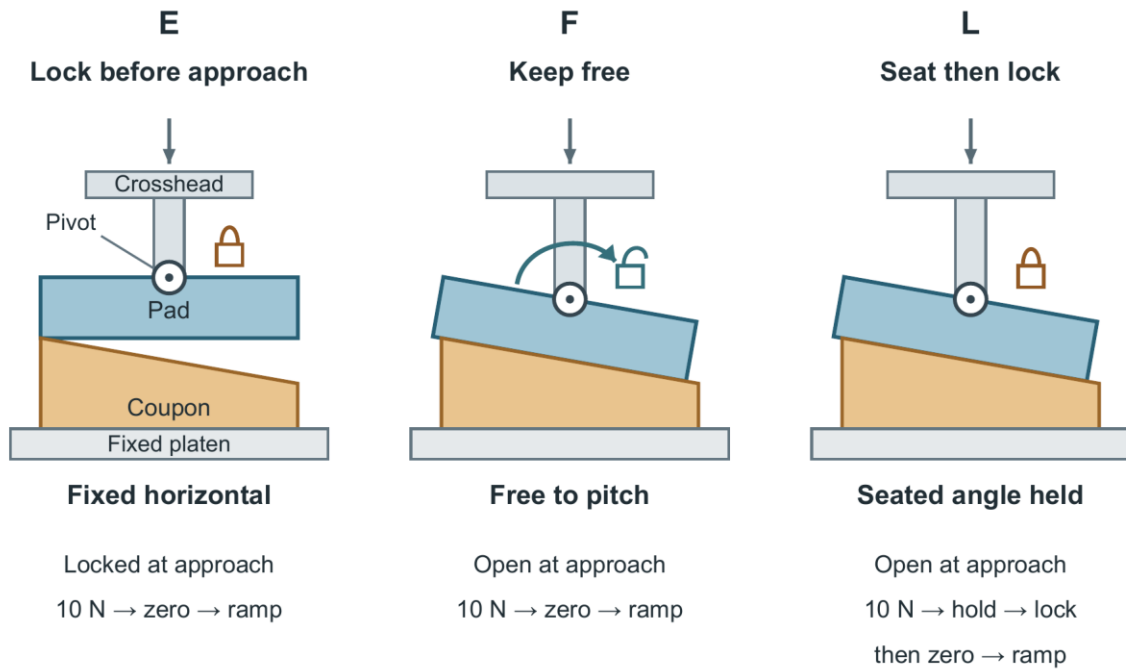
Results

All records are listed in Table 1. The pitch-slope comparison makes the tradeoff clearest. E gives 7.2% force CV, whereas F and L give 2.1% and 2.3%. Thus the free approach is associated with lower reseating variation. Keeping the clamp open through the ramp, however, gives 125 micrometres of drift in F; L gives 19, compared with 12 in E. L takes 32 seconds to seat, versus 18 for E and 17 for F. Locking after seating combines two favorable observations at a preparation-time cost; the records do not prove a uniform pressure field or an independent benefit of the acknowledgement step.

Table 1. Constructed repeated-mounting records.

Condition	Mode	Reseatings	Force CV (%)	Drift (μm)	Seating (s)
flat	E	8	1.3	14	18
flat	F	8	1.4	90	17
flat	L	8	1.3	16	31
pitch_1deg	E	8	7.2	12	18
pitch_1deg	F	8	2.1	125	17
pitch_1deg	L	8	2.3	19	32
roll_1deg	E	8	7.0	14	18
roll_1deg	F	8	6.8	126	17
roll_1deg	L	8	6.9	20	32

On the flat coupon, E and L both give a force CV of 1.3%, while F gives 1.4%. L adds 13 seconds over E and has slightly greater drift, 16 versus 14 micrometres. There is no demonstrated reason to add the hold-and-lock step for flat seating in this record. On the roll-slope coupon, force CV remains 7.0%, 6.8% and 6.9% for E, F and L. L reduces drift from F's 126 to 20 micrometres, but cannot provide the missing rotational freedom and takes 32 rather than 17 seconds. Low drift alone is therefore insufficient to judge successful seating. Across conditions, E retains low drift of 12-14 micrometres; F reaches 90-126, while L remains at 16-20. These are comparisons of complete procedures on one fixture, without uncertainty intervals to resolve small differences.



All ramps: 0.20 mm • pitch view • single axis • schematic DEMO

Figure 1. Clamp timing on the same single-axis fixture. Side views illustrate a pitch-tilted coupon at the start of the ramp: E retains a horizontal pad; F and L can settle to the slope, but only L is then locked. Closed and open lock symbols denote clamp state, not internal geometry. Downward arrows indicate loading; the curved arrow indicates permitted pitch rotation. The slope and E gap are exaggerated for visibility. The pin is normal to the page. The roll condition cannot be aligned by this axis and remains in Table 1. Poses and dimensions are schematic, not measured contact maps.

Conclusion

These constructed records favor separating seating from angular restraint when the coupon slope lies in the permitted pitch direction and lower drift justifies longer preparation. They do not support routinely choosing L for flat coupons or correcting a roll slope with one pivot. The procedural comparison establishes neither pressure uniformity nor unbiased material properties. Physical replication, randomized mode order and direct contact or angle measurements would be needed to test those broader claims.

Sources

[1] Teaching implementation notes and mode definitions. [2] Constructed records